SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SECURITY INCIDENT MAPPING SYSTEM (SIM)

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Security remains one of the most important concerns within educational institutions. Universities and colleges accommodate thousands of students, academic staff, non-academic staff, visitors, and residents every day. As a result, incidents such as theft, harassment, assault, vandalism, suspicious activities, and other security-related issues may occur both on and around campus. These incidents can negatively affect the safety of individuals, disrupt academic activities, and create fear within the university community.

In many institutions, security incidents are reported through informal methods such as verbal communication, phone calls, handwritten complaints, or personal visits to the security office. These traditional methods often result in delays, incomplete documentation, and poor communication between affected individuals and campus security personnel. Furthermore, there is usually no centralized platform where members of the campus community can easily access safety information or submit incident reports.

The Security Incident Mapping System (SIM) is proposed as a web-based solution that demonstrates how technology can improve campus security communication. The system provides an organized platform where students, staff, landlords, and campus security personnel can access safety information and submit reports of security incidents. Although the developed website is a static prototype created for academic purposes, it demonstrates the concept of an efficient digital reporting platform that can be expanded into a fully functional application in the future.

The development of SIM aligns with the growing adoption of information technology in solving real-world problems. By providing a user-friendly interface for reporting incidents and viewing security information, the system contributes to creating a safer campus environment while encouraging active participation from members of the university community.

1.2 Problem Statement

Campus security challenges continue to increase as educational institutions grow in population and physical size. Many incidents remain unreported because students and other members of the campus community do not have access to a convenient reporting platform. Existing reporting methods are often slow, inefficient, and difficult to monitor.

The absence of a centralized reporting system leads to delayed responses by security personnel, poor communication among stakeholders, inadequate documentation of security incidents, and reduced awareness of safety issues across the campus community. These limitations make it difficult for security personnel to identify recurring security concerns and develop effective preventive measures.

The Security Incident Mapping System was therefore developed to address these problems by providing a centralized interface through which security-related information can be presented and incident reports can be submitted in a structured and user-friendly manner.

1.3 Aim and Objectives

Aim:

The primary aim of this project is to develop a web-based prototype known as the Security Incident Mapping System (SIM) that demonstrates how technology can improve the reporting and awareness of security incidents within a university environment.

Objectives

The objectives of the project are to:

Design a modern and responsive website for campus security awareness.

Provide a structured incident reporting form.

Improve communication between students, staff, landlords, and campus security personnel.

Increase awareness of security issues within the campus community.

Present sample security reports for demonstration purposes.

Provide emergency contact information for campus users.

Demonstrate the application of HTML, CSS, and JavaScript in developing a web-based information system.

1.4 Scope of the System

The Security Incident Mapping System focuses on providing a frontend prototype of a campus security reporting website. The project demonstrates the appearance and expected functionality of a complete security reporting system without implementing backend technologies or databases.

The website consists of several sections, including the Home page, Dashboard, Report Incident page, Reports page, About page, and Contact page. Users can navigate through these sections, complete an incident reporting form, view sample reports, and access security-related information.

Since the project is intended for academic purposes, submitted reports are not permanently stored, user authentication is not implemented, and no server-side processing is performed. The system serves as a proof of concept that illustrates how a digital security reporting platform could function in a real-world environment.

1.5 Significance of the Study

The Security Incident Mapping System provides several important benefits to both the academic community and future system developers.

For students, the system demonstrates a simple method of reporting security incidents and accessing safety information. For staff members, it illustrates how security communication can be improved through digital technologies. Landlords can also benefit by understanding how incidents occurring around student hostels can be reported and monitored.

From an academic perspective, the project demonstrates practical knowledge of software engineering principles, web development, user interface design, and requirements analysis. It also provides a foundation upon which future developers can build a fully functional security management system using backend technologies and databases.

Furthermore, the project encourages the integration of technology into campus safety management and highlights the importance of effective communication in promoting a secure learning environment.

CHAPTER TWO

SYSTEM ANALYSIS

2.1 Existing System

The existing methods used for reporting security incidents in many institutions are largely manual. Students typically report incidents verbally to security personnel or through written complaints. Some institutions also rely on telephone calls or informal communication channels.

These methods present several limitations, including delayed reporting, poor documentation, lack of centralized information storage, and difficulty in monitoring recurring security incidents. Additionally, members of the campus community may not always know where or how to report security concerns, resulting in many incidents remaining undocumented.

Because information is scattered across different communication channels, security personnel may experience challenges in responding effectively and identifying patterns of criminal activities.

2.2 Proposed System

The proposed Security Incident Mapping System provides a centralized web-based platform that demonstrates how campus security information can be organized within a single interface.

The system contains sections for reporting incidents, viewing sample reports, learning about campus security, and accessing emergency contact information. Through a clean and user-friendly interface, users can easily navigate the website and understand how a digital reporting system operates.

Although the current implementation is static, the design provides a foundation for future enhancements such as user authentication, database integration, GPS-based incident mapping, email notifications, and real-time security alerts.

2.3 Functional Requirements

The system shall provide the following functions:

Display the homepage containing information about the Security Incident Mapping System.

Allow users to navigate between different pages of the website.

Display an incident reporting form.

Accept user input through the reporting form.

Display a confirmation message after a report is submitted.

Present sample security incident reports.

Display emergency contact information.

Provide information about the objectives and benefits of the system.

Support responsive layouts for different screen sizes.

Provide intuitive navigation for all users.

2.4 Non-Functional Requirements

The system should satisfy the following quality requirements:

Usability

The website should be easy to understand and simple to navigate for all users regardless of technical experience.

Performance

Pages should load quickly and respond efficiently to user interactions.

Reliability

The website should operate correctly on supported browsers without unexpected errors.

Maintainability

The source code should be well organized and documented to simplify future updates.

Compatibility

The system should function correctly on modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.

Responsiveness

The interface should automatically adjust to desktops, laptops, tablets, and smartphones.

Security

Although the prototype does not include backend security mechanisms, the design should encourage secure software development practices for future implementation.

2.5 Users of the System

The intended users of the Security Incident Mapping System include:

Students

Students can access security information, view campus safety updates, and complete the incident reporting form whenever security concerns arise.

Staff

Academic and non-academic staff can report security incidents affecting campus activities and promote awareness among students.

Landlords

Landlords can report incidents occurring around student residences and contribute to improving the security of off-campus accommodation.

Campus Security Personnel

Campus security personnel can use the information provided through the system to improve monitoring, identify security trends, and respond more effectively to reported incidents.

CONCLUSION

The Security Incident Mapping System (SIM) successfully demonstrates how a web-based platform can improve communication regarding campus security. By providing a centralized interface for reporting incidents, viewing safety information, and accessing emergency contacts, the system illustrates the potential benefits of digital solutions in enhancing campus safety.

Although the current implementation is a static website developed using HTML, CSS, and JavaScript, it effectively demonstrates the core concepts of a modern security reporting platform. Future versions of the system can be enhanced by integrating databases, user authentication, interactive maps, notification systems, and real-time reporting capabilities.

Overall, the project fulfills its objective of presenting a simple, user-friendly, and professional prototype that reflects the principles of software engineering while addressing an important real-world challenge within educational institutions.